

Diets high in palmitic acid (16:0), lauric and myristic acids (12:0 + 14:0), or oleic acid (18:1) do not alter postprandial or fasting plasma homoc...



BACKGROUND: Dietary fat type is known to modulate the plasma lipid profile, but its effects on plasma homocysteine and inflammatory markers are unclear. **OBJECTIVE:** We investigated the effects of high-protein Malaysian diets prepared with palm olein, coconut oil (CO), or virgin olive oil on plasma homocysteine and selected markers of inflammation and cardiovascular disease (CVD) in healthy adults. **DESIGN:** A randomized-crossover intervention with 3 dietary sequences of 5 wk each was conducted in 45 healthy subjects. The 3 test fats, namely palmitic acid (16:0)-rich palm olein (PO), lauric and myristic acid (12:0 + 14:0)-rich CO, and oleic acid (18:1)-rich virgin olive oil (OO), were incorporated at two-thirds of 30% fat calories into high-protein Malaysian diets. **RESULTS:** No significant differences were observed in the effects of the 3 diets on plasma total homocysteine (tHcy) and the inflammatory markers TNF- α , IL-1 β , IL-6, and IL-8, high-sensitivity C-reactive protein, and interferon- γ . Diets prepared with PO and OO had comparable nonhypercholesterolemic effects; the postprandial total cholesterol for both diets and all fasting lipid indexes for the OO diet were significantly lower ($P < 0.05$) than for the CO diet. Unlike the PO and OO diets, the CO diet was shown to decrease postprandial lipoprotein(a). **CONCLUSION:** Diets that were rich in saturated fatty acids prepared with either PO or CO, and an OO diet that was high in oleic acid, did not alter postprandial or fasting plasma concentrations of tHcy and selected inflammatory markers. This trial was registered at clinicaltrials.gov as NCT00941837.